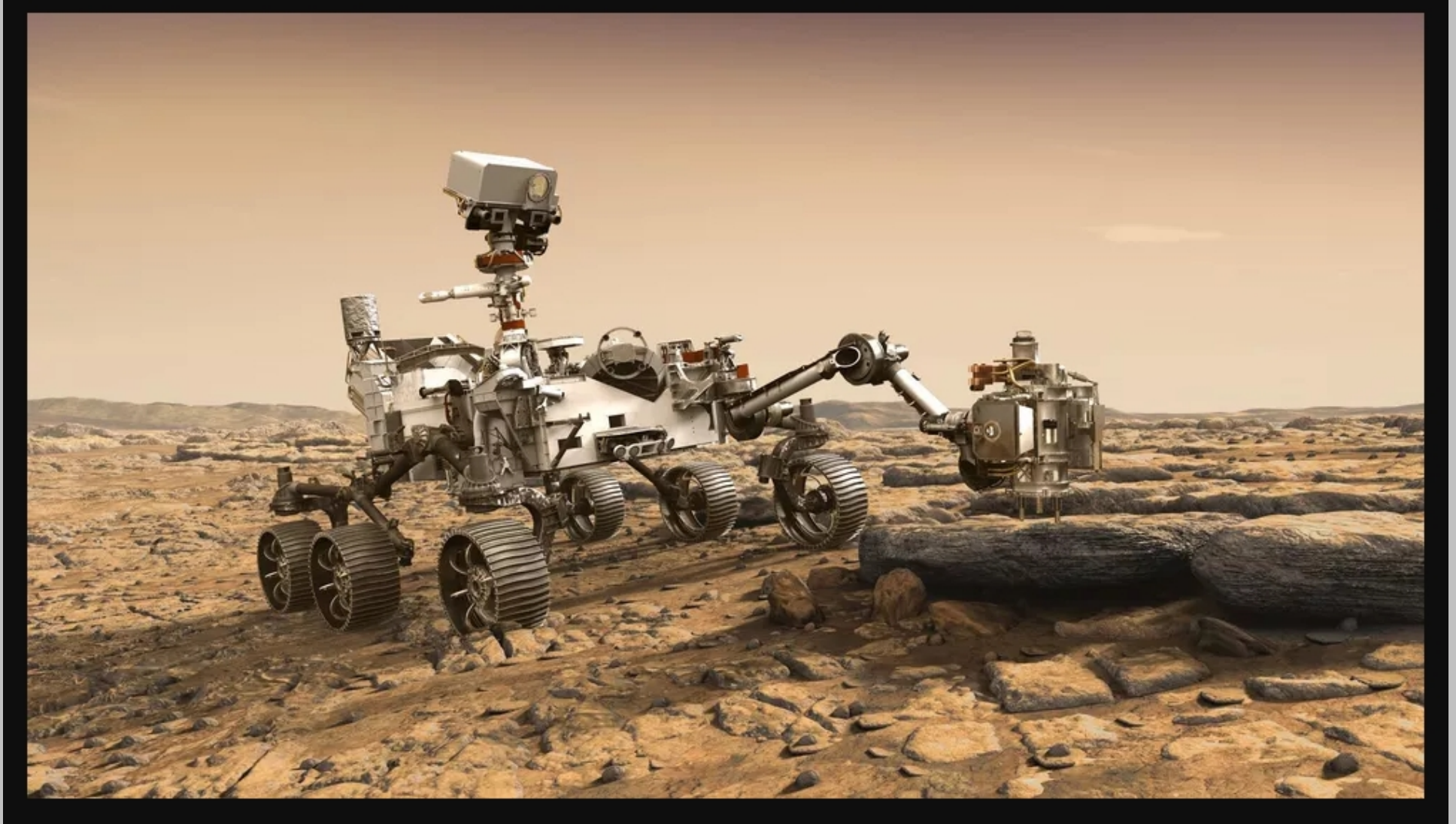


Planetary Astronomy

week 1



PHYS 1051

Dr. Jason Pinkney

Goals for this course

- 1) Obtain knowledge of astronomy.**
- 2) Recognize some of the laws of physics inherent in astronomy.**
- 3) Develop skills related to astronomy such as problem solving, math, and observing.**
- 4) Learn about science and how it differs from pseudosciences and other belief systems.**
- 5) Expand your personal “theory of everything” - your cosmology.**

Week 1 of Stellar and Galactic Astronomy

**View the film “*Powers of 10 A Film About the Relative Sizes of Things*”
1977**

by Charles and Ray Eames

Narrated by Phillip Morrison

Pre-Questions

- 1) Note the largest scale achieved.**
- 2) Note the smallest scale achieved.**
- 3) In which powers of 10 do we find great “voids” where nothing new enters the view?**

Also consider ...

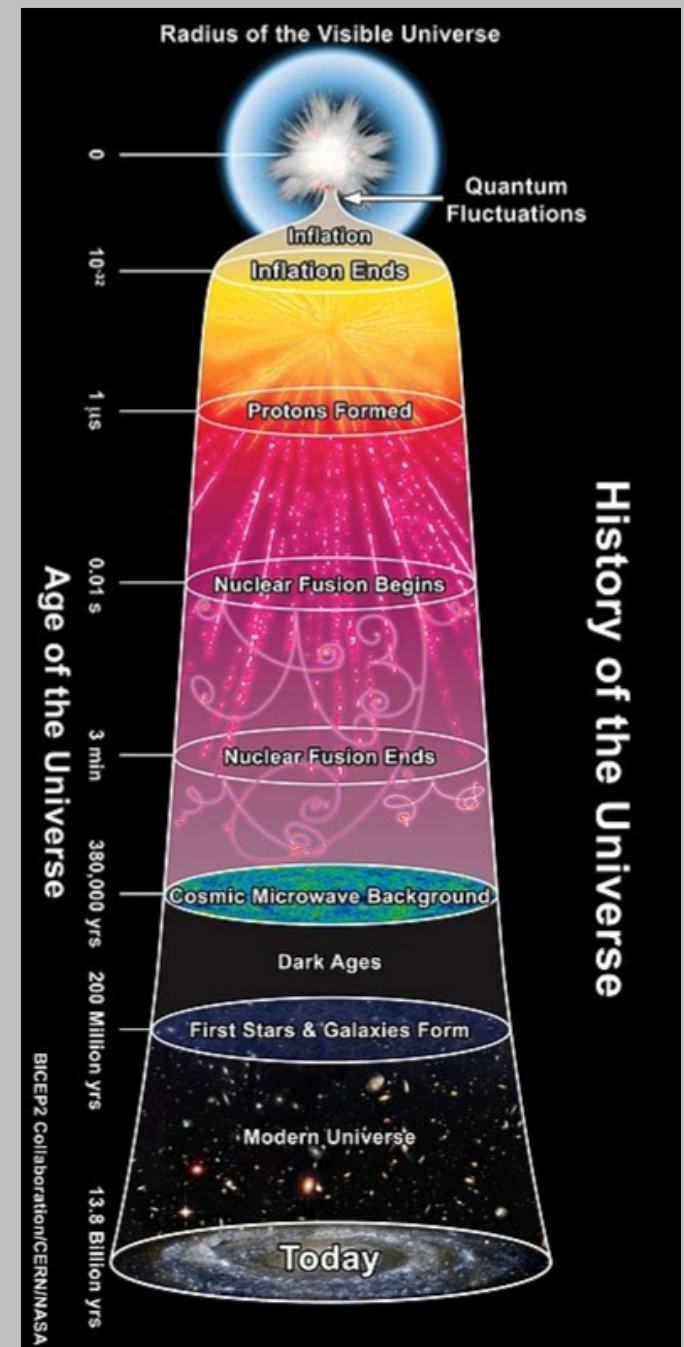
- 4) In every 10 seconds, the view expanded by a factor of 10. Could this entire journey be experienced while flying in a rocket ship?**

Why do we need powers of 10?

Many quantities have a vast range of values in Astronomy ...

- 1) *Distances*. 10^{-16} to 10^{24} meters (actually $>10^{26}$) for the scale of an atomic nucleus compared to the scale of largest structures in the universe.
- 2) *Time*. 10^{-46} second to 10^{16} seconds (10^9 yrs) for the Planck time to the age of the universe.
- 3) *Masses*: 10^{-31} kg (electrons) to 10^{45} kg (clusters of galaxies)
- 4) *Energies*: and 10^{-19} Joules for H-alpha photon to 10^{39} Joules for Gamma-Ray Bursts.
- 5) *Speeds*: continental drift (cm/yr) to the speed of light 300,000,000 m/s.

Ex) Without powers of 10, mass of Earth (in kg):
 $10^{25} = 10,000,000,000,000,000,000,000,000$



And that's why we use “powers of 10” -- to make them more manageable!

Understanding Scientific Notation, Powers of 10, and Orders of Magnitude

Scientific Notation: a way of writing a number in which the decimal point is placed to the right of the most significant digit, and this is multiplied by 10^P where P =an integer (the exponent, or “power of 10”)

Exponential Notation Format: **Coefficient** X **Base** ^{exponent}
(where Base=10)

Example: $58400 = 5.84 \times 10^4$

Example: $0.01093 = 1.093 \times 10^{-2}$

Example: The average Earth-Sun distance is 93,000,000 miles or 9.3×10^7 miles.

Power of 10: the exponent of a number expressed in scientific notation, rounded up or down depending on the coefficient.

Example: $5.84 \times 10^4 = 10^{4.7664} \sim 10^5 = 10 \times 10 \times 10 \times 10 \times 10 = 100,000$.

Example of usage: The distance to the Sun from the Earth is about 10^8 miles.
Thus, the Earth-Sun distance is 8 powers of 10 greater than a mile.

Order of magnitude: the “order of magnitude” of a number is the same thing as a number's integer “power of ten”, it is just used differently in sentences.

Example: “The Earth-Sun distance is 8 orders of magnitude larger than a mile.”)

Example: “If you thought the US population was 3 million, you were off by 2 orders of magnitude.”

Understanding Powers of 10, orders of magnitude, and Scientific Notation

Rounding to the nearest power of 10.

Previous example: $5.84 \times 10^4 = 10^{4.7664} \sim 10^5$. But what if we had ...

Example: $4.84 \times 10^4 = 10^{4.6848} \sim 10^5$.

Example: $3.84 \times 10^4 = 10^{4.5843} \sim 10^5$. Perhaps if the exponent dropped below 4.5 ...

Example: $2.84 \times 10^4 = 10^{4.4533} \sim 10^4$. Finally, we don't round up!

For which coefficient will the exponent be exactly 4.5? Answer: **3.162278** ($=\sqrt{10}$)

Example: $3.1623 \times 10^4 \sim 10^5$.

Example: $3.1622 \times 10^4 \sim 10^4$.

Try these:

Example: $9.99 \times 10^2 \sim 10^3$

Example: $9.9 \times 10^{-2} \sim 10^{-1}$.

Example: $5.1 \times 10^{-4} \sim 10^{-3}$.

Example: $3.10 \times 10^6 \sim 10^6$.

Example: $3.20 \times 10^9 \sim 10^{10}$.

Example: 401,000 $\sim 10^6$.

Example: 301,000 $\sim 10^5$.

Example: 73,162,055,319 $\sim 10^{11}$.

Why do we need these new ways of expressing numbers?

1) To compress long numbers.

Example: mass of the Sun in kilograms:

$$1 M_{\odot} = 2.0 \times 10^{30} \text{ kg (sci notation)}$$

$$1 M_{\odot} \sim 10^{30} \text{ kg (nearest power of 10)}$$

(Now try writing this number as a 2 with 30 zeros!)

2) To simplify multiplication and division.

Example: if the Earth's mass is 10^{25} kg, how many Earth masses go into the Sun?

$$M_{\odot}/M_{\oplus} = 10^{30} \div 10^{25} = 10^{30-25} = 10^5 \quad (\text{more precisely: } 3.3 \times 10^5 \sim 10^6)$$

Example: if there are 86400 seconds per day, and 365 days in a year, roughly how many seconds are in a calendar year?

$$10^5 * 10^3 = 10^{5+3} = 10^8 \quad (\text{more precisely: } 31,536,000 \sim 3.15 \times 10^7 \text{s})$$

$$10^5 * 10^2 = 10^{5+2} = 10^7 \quad (\text{to avoid rounding up twice})$$

Simplified multiplication and division allows easy rough estimates called “order of magnitude calculations” or “back of the envelope calculations”.

With powers of 10, division becomes subtraction and multiplication becomes addition.

Get some **Practice on Powers of 10 !** (See worksheet)

Planetary Astro OUTLINE

Week 2, Day 1

Today:

Review Pre-course survey

Instructions for “Practice on Powers of 10”

Prefixes and new units – other ways to deal with extreme numbers

Side notes on astronomical distances / sizes

The “Naked-Eye Universe”

Top Ten brightest objects

Constellations vs Asterisms

Homework:

Ch. 1 Disc 2-9 Due Friday

Ch. 1 Read 1.1-1.4

Worksheet on Pow. Of 10 Due Wed

Install Stellarium. Set Latitude = $40^{\circ} 46' 4.45''$ N (ONU Observatory)

Longitude = $83^{\circ} 50' 22.92''$ W

Notes: Pre-course survey average = 9.3/15

QUIZ 1 on Friday. See Review Questions under “NEW STUFF”.

Show me your notetaking software if you want to use laptop/tablet in class.

Last day to add is Tuesday, 9/1/26.

Other ways to make large numbers manageable

1) Use prefixes

small: deci, centi, milli, micro, nano, pico, femto, atto, zepto, yocto

10 to the: -1 -2 -3 -6 -9 -12 -15 -18 -21 -24

large: deka, hekto, kilo, mega, giga, tera, peta, exa, zeta, yotta

10 to the: 1 2 3 6 9 12 15 18 21 24

Example) What is a convenient unit for 10^{-6} seconds? Ans: a microsecond ($1 \mu\text{s}$).

2) Invent new units

In astronomy we have ... (red ones are new units)

a) The “solar mass”, $1 M_{\odot} = 2 \times 10^{30} \text{ kg}$

b) The “**astronomical unit**”, $1 \text{ AU} = 1.5 \times 10^8 \text{ km}$, 93,000,000 miles.

The average distance between the Earth and Sun.

c) The **Light year**, $1 \text{ LY} = 9.5 \times 10^{12} \text{ km}$

The distance light travels through space in a year.

Good for distances between stars.

d) The **parsec**, $1 \text{ pc} = 3.1 \times 10^{13} \text{ km}$.

The distance one must be from the Solar system so that the Earth-Sun separation appears to be 1 arcsecond.

Good for distances between stars.

e) The **kiloparsec**, $1 \text{ kpc} = 1000 \text{ pc}$

Good for distances inside a galaxy

f) The **megaparsec**, $1 \text{ Mpc} = 1,000,000 \text{ pc}$

Good for distances between galaxies, clusters, superclusters.

Side notes on astronomical distances / sizes

1) The magic number, 110.

110 (roughly) comes up many times in distance ratios.

$$110 = \text{DiamSun}/\text{DiamE} = \text{distSun}/\text{DiamSun} = \text{distMoon}/\text{DiamMoon}$$

2) The AU and Light Year.

There are 63,000 AU in 1 LY.

Coincidentally, there are 63,000 inches in a mile!

3) The distance to the Moon is 240,000 miles. A good car typically lasts about 240,000 miles. So you *might* be able to drive to the Moon if there were a direct route from Earth!

4) The ratio 400.

$$400 = \text{DistSun}/\text{distMoon} = \text{DiamSun}/\text{diamMoon}$$

Because of this coincidence the Sun and Moon subtend about the same angle in the sky ($\frac{1}{2}$ degree) and we can observe both total and annular solar eclipses.

The Naked – Eye Universe

Things that we can see with the naked eye

1. Sun
2. Moon
3. 5 planets (+Uranus, just visible)
4. 6500 stars (down to +6.0 mag)
5. 3 galaxies (M31,LMC,SMC. Some can see M33)
6. Comets
7. Supernovae, novae
8. Meteors (in our atmosphere)
9. Aurora (in our atmosphere)

The Naked – Eye Universe

The Top Ten Brightest Objects in the Sky

- 1. Sun**
- 2. Moon**
- 3. Venus**
- 4. Mars**
- 5. Jupiter**
- 6. Mercury**
- 7. Sirius**
- 8. Saturn**
- 9. Canopus (in Carina, Southern Hem)**
- 10. Alpha Centauri (Rigel Kentaurus)**

Arcturus, Vega, and Capella are almost a tie for 11th!

The Naked – Eye Universe

Constellations and Asterisms

Constellation: a designated region in the sky containing one or more historical star patterns. **Regions defined by the IAU.**

Asterism: a recognizable pattern of stars.

Ex) Orion, the hunter. Constellation AND asterism

Ex) Taurus, the bull.

Asterisms: the bull, the Pleiades; the Hyades

Ex) Ursa Major (the great bear)

Asterisms: the great bear, the Big Dipper; La Cassarole, pointer stars

Ex) The Summer Triangle – large asterism

Ex) The Coathanger (Brocchi's cluster) - telescopic asterism

- * 88 total constellations

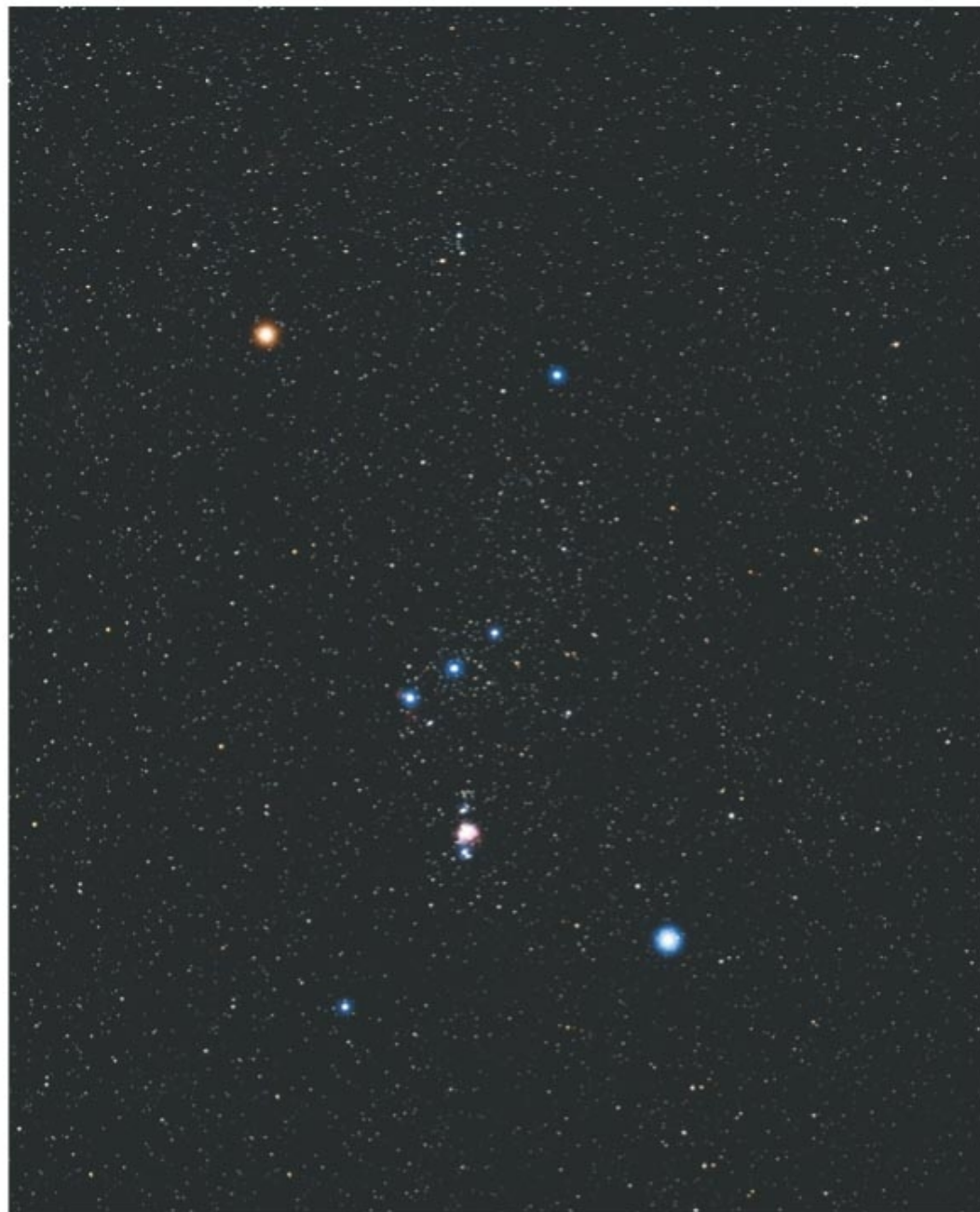
- * Many more than 88 asterisms!

- * Northern constellations named after Greek Mythological characters

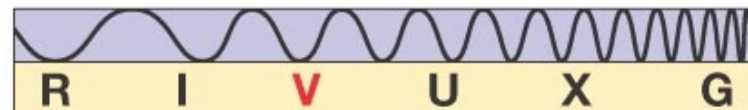
**Example:
Orion.**

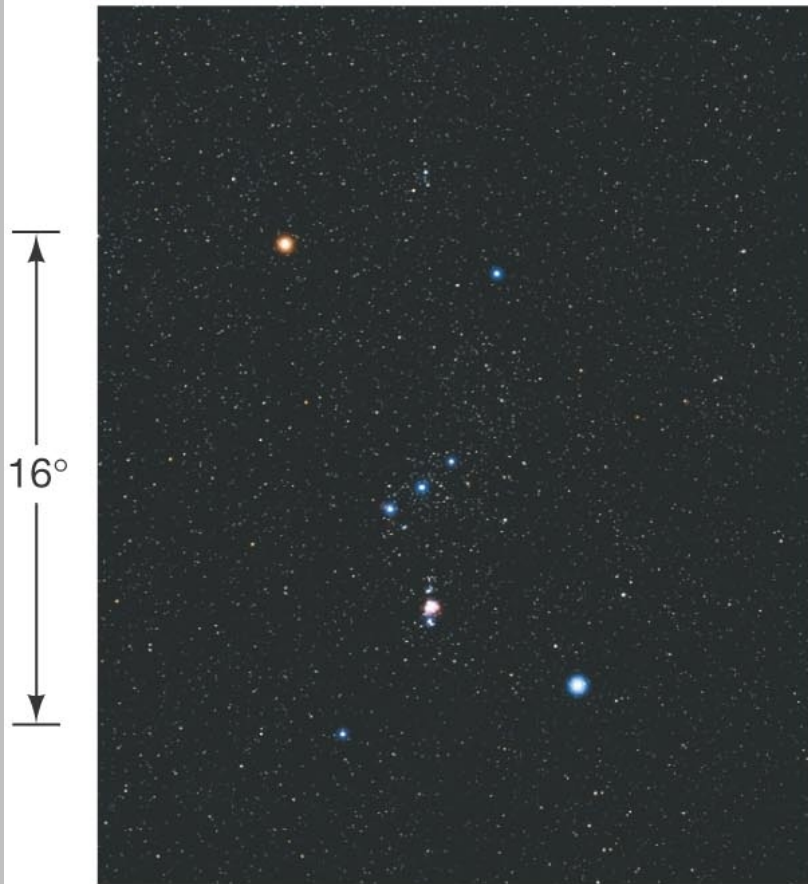
**An easily
recognized
constellation!**

16°

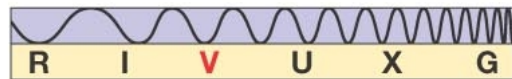


(a)

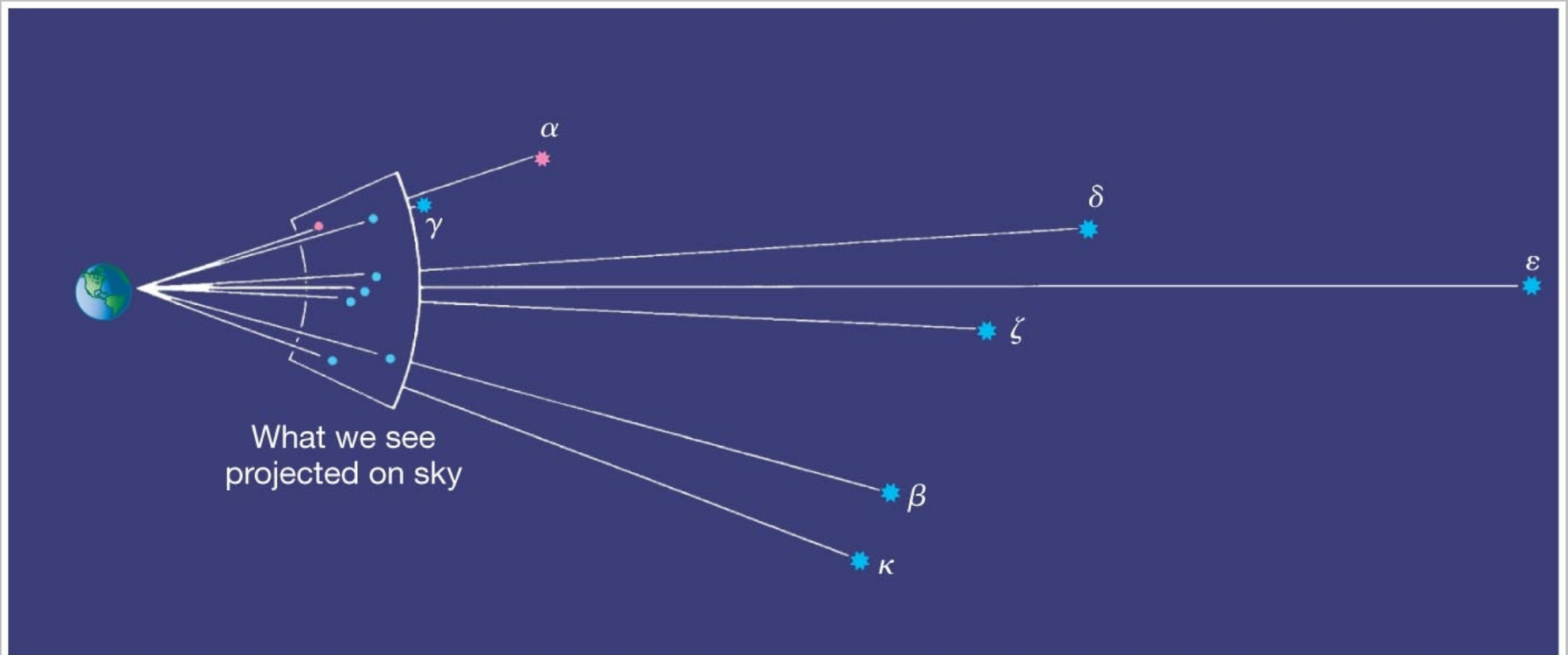




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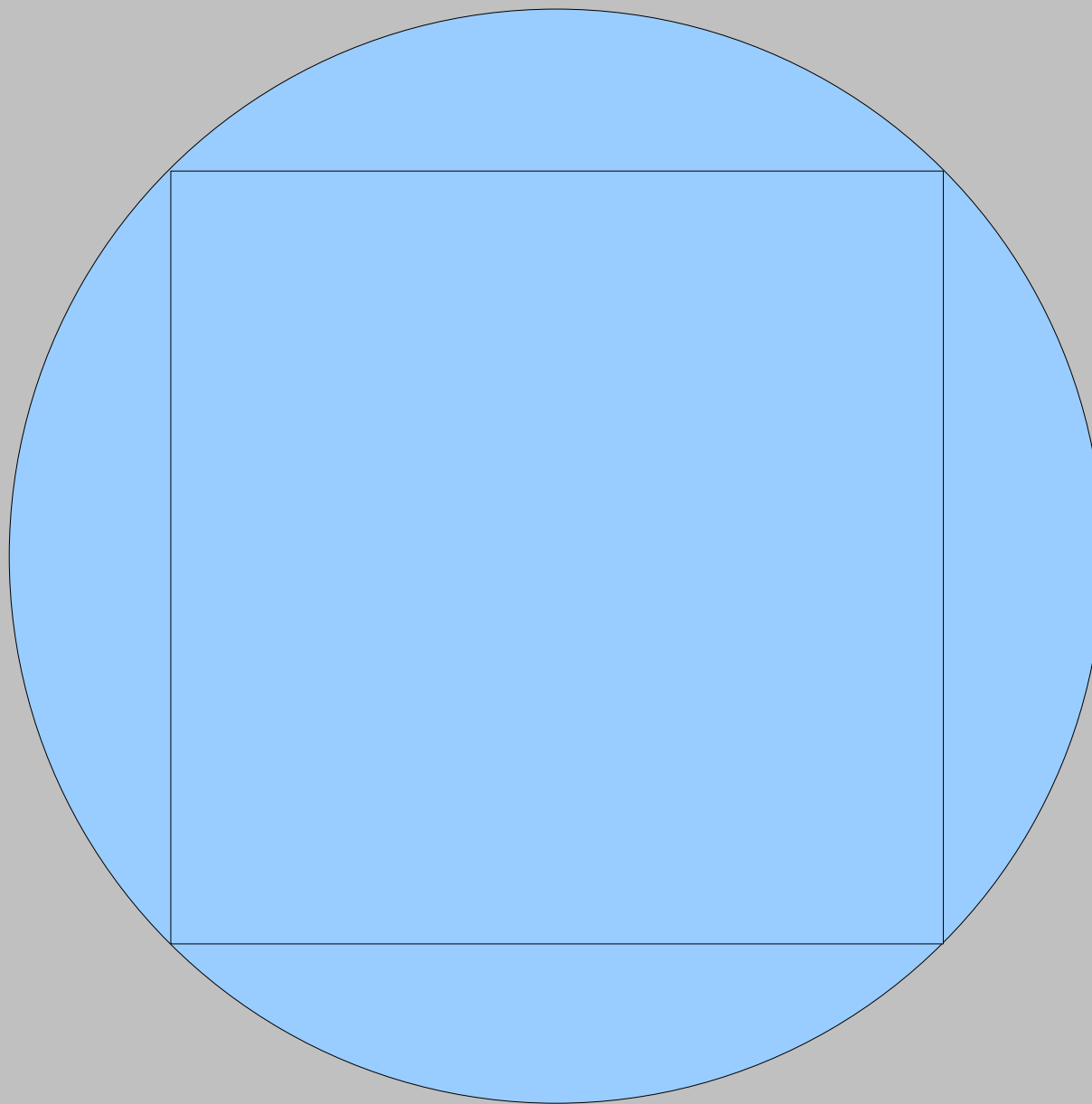
(b)





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Try a planetarium program like “Stellarium” to see the sky in motion.



Perfect Circle (for aspect ratio check)